(PV-STP-01) SwSTP

Document ID: `PV-STP-01`
Product: `Portview`
Document Status: `Released`

Document Overview

This document defines the unit-level software test procedure set for Portview.

Document notes:

- The current draft preserves the visible procedure grouping from the controlled test-procedure content.
- Execution details should remain aligned with `SV-04` and `SV-05`.

Document Approval

Prepared by

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Revision History

Rev.	Date	Description
`0.0`	`2024-01-17`	Initial procedure set prepared according to verification-item subdivision
`0.1`	`2025.07.24`	Added GUI translation test item

1. Purpose

This document defines the software unit test procedures used to demonstrate that Portview software functions satisfy their verification criteria.

2. References

This procedure set is based primarily on:

- `SwSRS for Portview`
- `SV-04` Software Verification Plan

3. Test Information

Field	Value
Tester name	`J. W. Lee`
Tests performed date	`2025.07.24`
System location	`Genoray testing platform`
Configuration	`Default`
Software baseline	`Portview 2.2.5.16`

4. Procedure Coverage

The current unit procedure set visibly covers the following items.

Authored Procedure ID	Coverage
`UTP-001`	Acquire intraoral sensor image
`UTP-002`	Acquire image from file
`UTP-003`	Acquire image from TWAIN device
`UTP-004`	Show status of device
`UTP-005`	Select device
`UTP-006`	Main-window behavior
`UTP-007`	Tooth map window
`UTP-008`	Compare image display
`UTP-009`	Pan behavior
`UTP-010`	Zoom behavior
`UTP-011`	Rotate behavior
`UTP-012`	Thumbnail viewer
`UTP-013`	Print image
`UTP-014`	Transmit to DICOM printer
`UTP-015`	Export image to file
`UTP-016`	Export images to CD
`UTP-017`	Send image to DICOM server
`UTP-018`	FMX continuous acquisition
`UTP-019`	Multi-sensor selection
`UTP-020`	Sensor status
`UTP-021`	USB integrity check

Authored Procedure ID	Coverage
`UTP-022`	Local-network connection check
`UTP-023`	Logging
`UTP-024`	Patient management
`UTP-025`	GUI translation

4.1 Unit Procedure Groups

The current unit procedure coverage can be organized into stable test groups.

Procedure Group	Included Procedures
Acquisition and device access	`UTP-001` to `UTP-005`, `UTP-018` to `UTP-022`
Viewer and image interaction	`UTP-006` to `UTP-012`
Output and communication	`UTP-013` to `UTP-017`
Administrative and support functions	`UTP-023` to `UTP-025`

4.2 Relationship To Requirements

The unit procedures are expected to verify software requirements from `SwSRS for Portview`, especially:

- acquisition and source-control behavior
- viewer behavior and interaction tools
- output and communication features
- logging and localization features

4.3 Detailed Procedure Summaries

Each item below captures the authored procedure intent at issue level. These summaries are intended to be specific enough to support detailed procedure authoring without forcing the full step transcript into the document body.

4.3.1 Acquisition And Device Access

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`UTP-001`	Portview with intraoral sensor and selected patient	Select device, open patient context, acquire sensor image	Sensor image is acquired and displayed under the selected patient
`UTP-002`	Portview with supported image files	Invoke import, choose file, load image	Selected file is imported without issue and displayed correctly
`UTP-003`	Portview with TWAIN device	Invoke TWAIN acquisition, select TWAIN device, acquire image	TWAIN image is acquired and displayed correctly
`UTP-004`	Portview with connectable device	Observe status before and after device connection	Device status changes correctly between disconnected and connected states
`UTP-005`	Portview with selectable device list	Open device-selection UI and change device	Selected device becomes active and viewer state follows the selected device type
`UTP-018`	Portview with FMX-capable acquisition setup	Enable FMX continuous acquisition and acquire in sequence	Continuous-acquisition workflow proceeds according to the selected FMX mode
`UTP-019`	Portview with multiple supported sensors	Open sensor-selection control and select alternate sensor	Selected sensor becomes active and acquisition path follows the chosen device
`UTP-020`	Portview with supported device connection state changes	Trigger device-state transitions	Sensor or device status is shown consistently for the current connection state
`UTP-021`	Installation or delivery media in controlled setup	Perform integrity check on delivered USB or installation media	Integrity-check result is available and unexpected corruption is detected

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`UTP-022`	Portview connected to local network path used by device workflow	Exercise network-dependent acquisition path	Network-based connectivity required for device workflow is available and stable

4.3.2 Viewer And Image Interaction

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`UTP-006`	Portview with patient records	Open patient list, select patient, observe main viewer	Main viewer displays the selected patient mount or current image set
`UTP-007`	Portview with patient images and tooth-map layout	Select patient image and change location through tooth map	Display changes to the image associated with the selected tooth location
`UTP-008`	Portview with patient images suitable for comparison	Enable compare command, choose images, apply comparison option	Comparison view is presented for the selected images
`UTP-009`	Portview with displayed image	Enable panning and drag image	Image position changes according to pan interaction
`UTP-010`	Portview with displayed image	Enable zoom and apply drag or zoom gesture	Image zoom level changes in and out as commanded
`UTP-011`	Portview with displayed image	Select rotate control and apply rotation or flip option	Image rotates or flips according to selected option
`UTP-012`	Portview with patient image set	Open patient and observe thumbnail area	Thumbnail viewer shows available images and selection state correctly

4.3.3 Output And Communication

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`UTP-013`	Portview with printable image	Invoke print command and complete print flow	Selected image is printed through the standard print path
`UTP-014`	Portview with configured DICOM printer	Invoke DICOM-print path and send selected image	Image is transmitted to DICOM printer without application error
`UTP-015`	Portview with exportable image	Choose export command, file format, and destination	Image export file is created successfully at target location
`UTP-016`	Portview with removable-media workflow	Invoke media export and complete target selection	Export to CD or equivalent media workflow completes successfully
`UTP-017`	Portview with reachable DICOM destination	Send selected image to DICOM server	DICOM transmission completes successfully or failure is detected and reported

4.3.4 Administrative And Support Functions

Authored Procedure ID	Setup	Core Procedure	Acceptance Criteria
`UTP-023`	Portview with audit-relevant user activity	Perform representative activity and inspect log output	Activity log records expected operations without corruption
`UTP-024`	Portview with patient-management permissions	Add, modify, retrieve, and manage patient records	Patient-management actions succeed and data remains consistent
`UTP-025`	Portview with multiple enabled language resources	Change GUI language and inspect visible UI labels	GUI text changes to the selected language without layout or content corruption

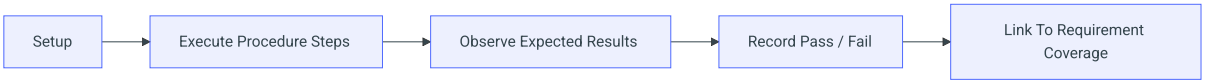
5. Procedure Structure

Each unit test procedure is expected to define:

- setup conditions
- ordered execution steps
- expected results
- pass or fail decision logic

5.1 Procedure Execution Model

Each unit procedure should be executable as a self-contained verification activity.



5.2 Evidence Expectations

Each executed unit procedure should produce or support:

- procedure execution date
- executor identification
- configuration under test
- pass or fail status
- deviation note if expected behavior is not observed

6. Current Test Environment Basis

The currently visible controlled procedure content identifies:

Environment Element	Current Basis
Test performer	`J. W. Lee`
Test date	`2025.07.24`
Test location	`Genoray testing platform`
Configuration	`Default`
Software under test	`Portview 2.2.5.16`

7. Relationship To The SV Set

This procedure set supports:

- planning content in `SV-04`
- executed evidence summarized in `SV-05`
- requirement coverage linkage in the Portview traceability matrix

8. Open Items

- `Conditional:` Add direct links to execution records if a separate unit-level result record is maintained.